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Roll No

EC-7003-CBGS

B.E. VII Semester

Examination, June 2020

Choice Based Grading System (CBGS)

Optical Communication

Time : Three Hours

Maximum Marks : 70

Note: i) Attempt any five questions.

ii) All questions carry equal marks.

iii) Make suitable assumptions wherever necessary.

1. a) Draw the block schematic of a typical digital optical fiber link. Explain in detail. 7
b) A typical relative refractive index difference for an optical fiber designed for long distance transmission is 1%. Estimate the Numerical Aperture and the solid acceptance angle in air for the fiber when the core index is 1.46. Further, calculate the critical angle at the core-cladding interface within the fiber. It may be assumed that the concepts of geometric optics hold for the fiber. 7
2. a) A multimode step index fiber with a core diameter of $80\text{ }\mu\text{m}$ and a relative index difference of 1.5% is operating at a wavelength of $0.85\text{ }\mu\text{m}$. If the core refractive index is 1.48, Estimate: 7
i) the normalized frequency for the fiber;
ii) the number of guided modes.
b) Compare and contrast the performance attributes, potential drawbacks and possible applications of index-guided PCFs (Photonic Crystal Fibers) and photonic bandgap fibers. 7

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3. a) Discuss the mechanism of optical feedback to provide oscillation and hence amplification within the laser. Indicate how this provides a distinctive spectral output from the device. 7
- b) A ruby laser contains a crystal of length 4 cm with a refractive index of 1.78. The peak emission wavelength from the device is $0.55 \mu\text{m}$. Determine the number of longitudinal modes and their frequency separation. 7
4. a) Briefly outline the advantages and drawbacks of the LED in comparison with the injection laser for use as a source in optical fiber communications. 6
- b) The light output $0.013P_{\text{int}}$ from the GaAs LED is coupled into a step index fiber with a numerical aperture of 0.2, a core refractive index of 1.4 and a diameter larger than the diameter of the device. Estimate: 8
 - i) The coupling efficiency into the fiber when the LED is in close proximity to the fiber core.
 - ii) The optical loss in decibels, relative to the power emitted from the LED, when coupling the light output into the fiber.
 - iii) The loss relative to the internally generated optical power in the device when coupling the light output into the fiber when there is a small air gap between the LED and the fiber core.
5. a) Explain the detection process in the $p-n$ photodiode. Compare this device with the $p-i-n$ photodiode. 6
- b) A photodiode has a quantum efficiency of 65% when photons of energy $1.5 \times 10^{-19} \text{ J}$ are incident upon it. 8
 - i) At what wavelength is the photodiode operating?
 - ii) Calculate the incident optical power required to obtain a photocurrent of $2.5 \mu\text{A}$ when the photodiode is operating as described above.

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6. a) Describe the eye pattern analysis for assessing the performance of a digital fiber optical link. 7
- b) Following are the parameters of a point-to-point optical link: 7
- i) Optical power launched: +2 dBm
 - ii) Sensitivity of detector: -22 dBm
 - iii) Source/detector connector loss : 1 dB
 - iv) Length of optical cable : 50 km
 - v) Cable attenuation : 0.3 dB/km
 - vi) Jumper cable loss : 2 dB
 - vii) Connector loss at each fiber joint : 1 dB
- (two at each transmitter and receiver end because of the jumper cables). Compute the power margin of the link using spread sheet method.
7. a) Using an energy band diagram, briefly discuss the mechanism for the provision of stimulated emission in the erbium-doped silica fiber amplifier. Name and describe a phenomenon occurring in this material system which creates a limitation to the optical gain that may be obtained from the device. 7
- b) An SOA operating at a signal wavelength of 1.55 μm produces a gain of 30 dB with an optical bandwidth of 1 THz. The device has a spontaneous emission factor of 4 and the mode number is equal to 2.2 when the net gain coefficient over the length of amplifier is 200. 7
- Determine:
- i) the length of the device
 - ii) the ASE noise signal

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8. Write notes on any two of the followings: 2×7=14
- i) Optical Time-Domain Reflectometer
 - ii) WDM
 - iii) Coherent Detection
 - iv) Avalanche Photodiode
